

# Sarah A. Carneiro, PhD.

AI Research Scientist | AI Research Engineer | Data Scientist

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## EDUCATION

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### Post-doctoral Researcher

*École polytechnique (LIX)*

Paris, France

2025 - Present

### Ph.D. in Signal, Image, and Automation

*Université Gustave Eiffel (UGE)*

Paris, France

2020 - 2024

### M.Sc. in Computer Science

*University of Campinas (UNICAMP)*

Campinas, Brazil

2020

## AI RESEARCH EXPERIENCE

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### Postdoctoral Researcher Scientist – Multimodal Bio Generative AI

*École Polytechnique – LIX, Paris, France*

2025–Present

- Research on **multimodal foundation models** for scientific and biological AI, integrating protein structures, sequences, language, and structured scientific information.
- Contributed to the development of **PPI2Text**, a multimodal foundation-model framework for translating protein–protein interaction structures into interpretable biological knowledge.
- Worked on **Prot2Text-V2**, a multimodal protein representation model aligning protein sequences with natural language through contrastive learning.
- Designed and evaluated large-scale experiments, including ablation studies and benchmarking for protein function prediction under low-homology biological settings, with emphasis on robustness and generalization.
- Led the **evaluation and benchmarking of LLMs** team within the **OPENLLM France** consortium, covering scientific, open-source, and sovereign AI systems.
- **Represented the lab** in research consortium and funding meetings, including **OPENLLM**, contributing to discussions on collaborative research directions and funding opportunities.
- **Wrote research proposals** for DaSciM’s funding activities, including submissions to **Hi! PARIS** and **ANR** funding programmes.

### PhD Research Scientist – Deep Learning Architectures

*IFP Energies Nouvelles/ Université Gustave Eiffel, Paris, France*

2020–2024

- Designed CNN, RNN, and GNN architectures for structured and spatio-temporal prediction tasks.
- Integrated structured and geometric inductive biases into RNN-based architectures.
- Built scalable experimental pipelines and reproducible benchmarking frameworks.
- Teaching university classes in deep learning, image analysis, AI, optimization, algorithms, and data structures.

### Research Scientist – Multimedia Deep Learning

*Semantix, Campinas, Brazil*

2017–2019

- Developed multi-stream deep neural network ensembles for computer vision human posing in video understanding tasks.

## TECHNICAL EXPERTISE

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- **Multimodal & Scientific AI:** Multimodal representation learning; vision-language models; cross-modal alignment and fusion; foundation models; scientific reasoning; biomolecular AI.
- **Generative AI & Machine Learning:** LLMs; instruction tuning and parameter-efficient adaptation; contrastive and generative learning; CNNs, GNNs, and temporal models; supervised and self-supervised learning.
- **Frameworks & Engineering:** Python; PyTorch; Hugging Face; Transformers; OpenCV; SciPy; Tensorflow; Java.
- **Languages:** Portuguese, English, French.

## OTHER ONGOING PROJECTS

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- **Agentic AI:** Co-supervision of a research project on agentic systems and constrained language-model generation.
- **LLM Evaluation & Quantization:** Supervision of a research project on benchmarking and evaluating large language models under quantization.
- **Computer Vision for Medical Imaging:** Supervision of a research project applying computer vision methods to medical image analysis.
- **Deepfake Detection:** Supervision of an undergraduate research project on computer vision methods for detecting AI-generated and manipulated images.
- **Plant Disease Detection:** Supervision of an undergraduate research project applying computer vision to automated plant disease classification.

## SELECTED PUBLICATIONS

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- Prot2Text-V2: Multimodal Protein-to-Text Generation for Function Prediction. (2025, Neural Information Processing Systems (**NeurIPS**)).
- The Masked Advantage: Uncovering Local-Language Access to Cultural Knowledge in LLMs. (2026, arXiv preprint).
- CARTE: A Benchmark for Mapping Language Model Knowledge Across France. (2026, arXiv preprint).
- PPI2Text: Captioning Protein-Protein Interactions with Coordinate-Aligned Pair-Map Decoding. (2026, arXiv preprint).
- Every Wrong Answer Counts: Option-Level Psychometrics for LLM Multiple-Choice Benchmarks. (2026, arXiv preprint).
- SWMLP: Shared Weight Multilayer Perceptron for Car Trajectory Speed Prediction Using Road Topographical Features. (2023, 8th International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS)).
- Graph-Based Supervoxel Computation from Iterative Spanning Forest. (2021, International Conference on Discrete Geometry and Mathematical Morphology).
- Fight Detection in Video Sequences Based on Multi-Stream Convolutional Neural Networks. (2019, 32nd SIBGRAPI Conference on Graphics, Patterns and Images).

## ACADEMIC SERVICE & OUTREACH

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- **Symposium on Generative AI for Biochemistry and Health** — Member of scientific committee. Supported the organization and promotion of the symposium by drafting promotional materials, and identifying and contacting academic and industry professionals to invite them as speakers.
- **DaSciM Lab** — Led the update of DaSciM's 2026 visual identity.